

Youssef Serag

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EDUCATION

University of South Florida

Bachelor of Science in Computer Science

Tampa, FL

Aug. 2023 – Dec 2026

EXPERIENCE

Amazon

Austin, TX

Software Development Engineer Intern

May 2025 – Aug. 2025

- Designed and implemented the **List Price Nudging** Tool, resolving a bug causing **duplicate seller emails** and addressing **14M** outdated prices, cutting incident email volume **from 350K to 25K**.
- Rewrote core **AWS Lambda functions in Java** to parse, validate, and route pricing data for automated seller notifications, replacing deprecated infrastructure.
- Architected and deployed AWS infrastructure** using CDK across development, beta, and production environments, ensuring scalability and consistency of cloud resources.
- Built a **CI/CD pipeline** integrating automated builds, JUnit/Mockito testing, deployments, and security validations.

Teaching Assistant – Data Structures

Tampa, FL

University of South Florida

Aug 2025 – Present

- Support a **140-student** course through twice-weekly **office hours**, exam proctoring, and grading, while explaining material one-on-one and advising students on their careers.

PROJECTS

PediaTrack – Clinician-Guided Patient Recovery Platform | *Java, Spring Boot, PostgreSQL, React, JUnit, Mockito*

- Designed and implemented a **Java Spring Boot backend** using layered architecture (controllers, services, repositories) for patient recovery data, treatment plans, and clinician-patient communication with **strict role-based access control**.
- Built structured recovery-tracking workflows with validated daily symptom check-ins and timeline views, using **rule-based threshold logic** to flag abnormal trends and **reduce unstructured after-hours communication**.
- Modeled and optimized a **relational PostgreSQL schema** with indexed clinical entities (patients, treatments, medications, updates), and exposed **versioned REST APIs** for future mobile clients.

PrepWise - AI Voice Interview Platform | *Next.js, React, VAPI AI, Firebase, TailwindCSS*

- Developed an AI-powered interview platform conducting **real-time voice interviews** using VAPI and delivering **Gemini-generated feedback** to simulate real recruiter interactions.
- Built a **full-stack system** with **Next.js App Router**, **Firebase Auth**, and **Firestore**, designed for **100+ concurrent sessions** with secure httpOnly cookies and role-based routes.
- Integrated **Google Gemini 2.0 Flash** to generate **customized interview questions** and feedback validated through **Zod schemas**, reducing malformed AI responses by **95%**.

Bellini Classes – Course Scheduling Platform | *Next.js 14, TypeScript, React, Node.js, Tailwind CSS*

- Built a full-stack scheduling platform for a college's scheduling committee, including CRUD operations, a data audit report with 9 validation rules, multi-field search, and instructor/room views.
- Designed a semester comparison feature comparing course offerings between any two semesters using a subject-and-course-number key instead of CRN, since CRNs are reissued each semester.
- Built an enrollment and waitlist analysis feature flagging high-demand classes (90%+ capacity or active waitlist), including a fallback for a source file missing enrollment data entirely.

The Last of Us: Legacy – Game | *Java, JUnit, Java Swing*

- Developed a 15x15 grid-based strategy game with multi-character mechanics (*Attack, Move, Collect, Cure, Special Abilities*) across **over 10,000 potential grid locations**.
- Built a responsive **Java Swing GUI** processing **50+ inputs per turn**, backed by **250+ JUnit tests** achieving **85% code coverage** across all core mechanics.

SKILLS & LEADERSHIP

Technical Skills: Java, Python, C/C++/C Sharp, JavaScript, TypeScript, SQL, Spring Boot, REST APIs, React, Next.js, Node.js, HTML, CSS, Tailwind CSS, PostgreSQL, MongoDB, Firebase, JUnit, Mockito, AWS (Lambda, S3, DynamoDB, SQS), CI/CD, Git, CloudWatch, Integration Testing, Object-Oriented Design (OOD)

Leadership: Tech Lead, Google Developers Student Club – Organized HackUSF (300+ participants) and led hands-on Java workshops to teach practical programming and system design to peers.